LAB WORKSHEET

Command Prompt Lab

Use the Windows Command Prompt to inspect a PC's network settings, test connectivity, query DNS and spot suspicious connections — then solve a practice support ticket.

Part 1 — Explore

Open the **Explore** tab. The monitor shows a Windows desktop with Command Prompt already open. Type each command and note what it tells you. Add `/?` to any command for its options.

Command	What it shows / does
`ipconfig` / `ipconfig /all`	IP address, subnet mask, default gateway; `/all` adds MAC, DHCP and DNS servers
`ipconfig /release`, `/renew`, `/flushdns`	give up / request a DHCP lease; clear the DNS cache
`ping 192.168.10.1`	tests reachability with ICMP echo; shows round-trip time and TTL. `-n 10` sends 10, `-t` runs until Ctrl+C
`tracert` <host>	lists every router (hop) on the path; `-d` skips name lookups
`nslookup` <name>	asks a DNS server for the IP address of a name
`netstat -ano`	active connections and listening ports, with the owning process ID (PID)
`arp -a`	the ARP cache: IP-to-MAC mappings learned on the LAN
`route print`	the PC's own routing table, including its default route
`tasklist` / `taskkill /PID n`	list running processes; end one by its PID
`whois` <IP or domain>	who an address or domain is registered to
`netsh interface ip show config`	interface settings; `netsh` can also change them

Part 2 — Practice task: the reception PC

Open the **Practice task** tab. You start on the reception PC's desktop. Read **IT_notes.txt**, then double-click the **Command Prompt** shortcut. Work through the objectives — they tick off as you go.

1. Record the PC's IP address, gateway and DNS server.
2. Prove the PC can reach its gateway and the internet by IP address.
3. Show that name resolution is failing, and find why.
4. Correct the DNS server and prove that names now resolve.
5. Find the suspicious outbound connection, identify the process behind it, look up who owns the remote address, and end the process.

Method: test from the bottom up. If pinging an IP works but a name fails, the problem is DNS — not cabling or routing.

Test log

Test	Expected	Actual	Action & re-test